

# 1 Introduction

BioPortal provides APIs to search biomedical terms in several ontologies. This study explores the question, “how consistent are the search results in Hybrid Search with those in BioPortal?” The ability of Hybrid Search to find relevant entities is assessed, assuming that the search results in BioPortal represent the ground-truth.

## 2 Objective

To evaluate precision and recall of Hybrid Search application using BioPortal as the ground-truth reference.

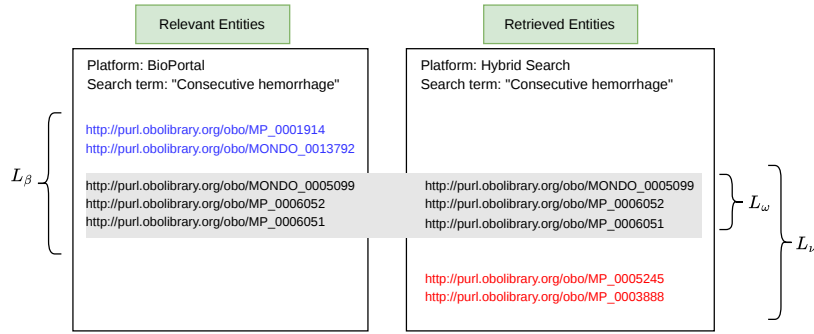


Figure 1: The search results in BioPortal and Hybrid Search platforms are shown. The common entities found in both platforms are highlighted by the grey box. Precision ( $L_\omega/L_\nu$ ) and Recall ( $L_\omega/L_\beta$ ) can be compared.

## 3 Approach

BioPortal APIs enable searching different types of resources. We develop the reference datasets (ground-truth) as follows.

- Identify target ontologies. E.g. MONDO, CHIRO, etc.
- Using BioPortal API, search for terms in the target ontologies.
- Extract the relevant fields and save the output (**golden** dataset).

Subsequently, the same queries used for curating the datasets from BioPortal are run in the Hybrid Search platform for comparison as follows.

- Using hybrid search APIs, run the same queries for finding entities in the target ontologies.
- Compare the results with the golden dataset.

## 4 Datasets

Descriptions of the datasets curated using BioPortal API are provided in Table 1. The number of items retrieved from BioPortal for each query was capped at  $k = 20$ . The value of  $k$  was arbitrarily picked, and can be changed as needed in the future.

Table 1: Dataset descriptions.

|                                                                                                                                                                                                                                                                    |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Dataset ID:</b> DS01                                                                                                                                                                                                                                            |
| <b>Filename:</b> bp_class_search_chiro_mondo_at_least_n_minus_one.jsonl                                                                                                                                                                                            |
| <b>Context:</b> Unspecified.                                                                                                                                                                                                                                       |
| <b>Description:</b> For a multi-word search query containing $N_{terms}$ unique terms, find entities where at least $N_{terms} - 1$ of those terms are found in the fields ‘definition’, ‘prefLabel’, and ‘synonym’ (accumulative, not necessarily in each field). |

## 5 Results

The performance evaluation of hybrid search platform started in different retrieval and reranker modes is summarized in Table 2. As described in Section 4, the value of  $k$  represents the maximum number of items curated for each query in the reference (BioPortal) dataset. The same value of  $k$  was used as *max\_results* parameter when calling the hybrid search endpoints. Let  $L_{dataset}$  denote the number of items in the dataset, Q05 and Q95 denote the 5-th and 95-th quantiles.

Table 2: Test results with  $L_{dataset} = 30$  and  $k = 20$ .

| Dataset ID | Retrieval mode | Reranker type | Results                                                                                                                             |
|------------|----------------|---------------|-------------------------------------------------------------------------------------------------------------------------------------|
| DS01       | Hybrid         | ensemble      | Mean precision@k=0.233<br>Q05 precision@k=0.0<br>Q95 precision@k=1.0<br>Mean recall@k=0.109<br>Q05 recall@k=0.0<br>Q95 recall@k=0.7 |
| DS01       | BM25-only      | biomedical    | Mean precision@k=0.217<br>Q05 precision@k=0.0<br>Q95 precision@k=1.0<br>Mean recall@k=0.101<br>Q05 recall@k=0.0<br>Q95 recall@k=0.7 |